STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 5: Work, Energy and Power

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

When is work zero with displacement? [1]
Kinetic energy? [1]
SI unit of power? [1]
Work-energy theorem? [1]
Spring energy? [1]
What is conserved in an isolated elastic collision? [1]
Near-earth gravitational potential energy? [1]
Work by centripetal force in uniform circular motion? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

10 N over 4 m along force: work? [2]
3 kg at 4 m/s: K? [2]
100 W for 20 s: energy? [2]
200 N perpendicular to 5 m displacement: work? [2]
k=200 N/m, spring stretched .1 m: energy? [2]
1 kg cart at 5 m/s sticks to stationary 4 kg: speed? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

2 kg speeds from 2 to 6 m/s: net work? [3]
5 kg falls 3 m, g=10: final speed? [3]
F=4x N from x=0 to 3 m: work? [3]
1000 kg car at 20 m/s stops in 50 m: W and force magnitude? [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

2 kg drops 5 m frictionlessly, g=10: speed; with 20 J loss? [4]
1 kg cart at 8 m/s sticks to 3 kg stationary: speed and energy loss? [4]

End of paper - check working, units, signs and significant figures.